

Experiment 7

AIM: Use Selenium to create browser-based tests

Step 1 Import Required Libraries

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from webdriver_manager.chrome import ChromeDriverManager
import time
```

Step 2 Start Browser

```
service = Service(ChromeDriverManager().install())
driver = webdriver.Chrome(service=service)
driver.get("https://the-internet.herokuapp.com/login")
driver.maximize_window()
```

Step 3 Wait for Page Elements

```
wait = WebDriverWait(driver, 10)
```

Step 4 Enter Login Credentials

```
username = wait.until(
    EC.presence_of_element_located((By.ID, "username")))
username.send_keys("tomsmith")
driver.find_element(By.ID, "password").send_keys("SuperSecretPassword!")
```

Step 5 Click Login Button

```
driver.find_element(By.CSS_SELECTOR, "button.radius").click()
```

Step 6 Verify Login Success

```
message = wait.until(
    EC.presence_of_element_located((By.ID, "flash")))
print("Login Message:", message.text)
```

Step 7 Logout

```
driver.find_element(By.CSS_SELECTOR, "a.button.secondary.radius").click()
```

Step 8 Close Browser

```
time.sleep(3)
driver.quit()
```

Command Prompt:

Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

```
C:\Users\kisha>mkdir selenium-lab
C:\Users\kisha>cd selenium-lab
```

```
C:\Users\kisha\selenium-lab>python --version
Python 3.12.10
```

```
C:\Users\kisha\selenium-lab>pip install selenium
Defaulting to user installation because normal site-packages is not writeable
Collecting selenium
  Downloading selenium-4.41.0-py3-none-any.whl.metadata (7.5 kB)
Collecting certifi>=2026.1.4 (from selenium)
  Downloading certifi-2026.2.25-py3-none-any.whl.metadata (2.5 kB)
Collecting trio<1.0,>=0.31.0 (from selenium)
  Downloading trio-0.33.0-py3-none-any.whl.metadata (8.5 kB)
Collecting trio-websocket<1.0,>=0.12.2 (from selenium)
  Downloading trio_websocket-0.12.2-py3-none-any.whl.metadata (5.1 kB)
Requirement already satisfied: typing_extensions<5.0,>=4.15.0 in
c:\users\kisha\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\loca
lcache\local-packages\python312\site-packages (from selenium) (4.15.0)
Collecting urllib3<3.0,>=2.6.3 (from urllib3[socks]<3.0,>=2.6.3->selenium)
  Downloading urllib3-2.6.3-py3-none-any.whl.metadata (6.9 kB)
Collecting websocket-client<2.0,>=1.8.0 (from selenium)
  Downloading websocket_client-1.9.0-py3-none-any.whl.metadata (8.3 kB)
Collecting attrs>=23.2.0 (from trio<1.0,>=0.31.0->selenium)
  Downloading attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)
Collecting sortedcontainers (from trio<1.0,>=0.31.0->selenium)
  Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)
Collecting idna (from trio<1.0,>=0.31.0->selenium)
  Downloading idna-3.11-py3-none-any.whl.metadata (8.4 kB)
Collecting outcome (from trio<1.0,>=0.31.0->selenium)
  Downloading outcome-1.3.0.post0-py2.py3-none-any.whl.metadata (2.6 kB)
```

Collecting sniffio>=1.3.0 (from trio<1.0,>=0.31.0->selenium)
Downloading sniffio-1.3.1-py3-none-any.whl.metadata (3.9 kB)
Collecting cffi>=1.14 (from trio<1.0,>=0.31.0->selenium)
Downloading cffi-2.0.0-cp312-cp312-win_amd64.whl.metadata (2.6 kB)
Collecting wsproto>=0.14 (from trio-websocket<1.0,>=0.12.2->selenium)
Downloading wsproto-1.3.2-py3-none-any.whl.metadata (5.2 kB)
Collecting pysocks!=1.5.7,<2.0,>=1.5.6 (from urllib3[socks]<3.0,>=2.6.3->selenium)
Downloading PySocks-1.7.1-py3-none-any.whl.metadata (13 kB)
Collecting pycparser (from cffi>=1.14->trio<1.0,>=0.31.0->selenium)
Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Collecting h11<1,>=0.16.0 (from wsproto>=0.14->trio-websocket<1.0,>=0.12.2->selenium)
Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Downloading selenium-4.41.0-py3-none-any.whl (9.5 MB)

9.5/9.5 MB 2.5 MB/s eta 0:00:00

Downloading certifi-2026.2.25-py3-none-any.whl (153 kB)
Downloading trio-0.33.0-py3-none-any.whl (510 kB)
Downloading trio_websocket-0.12.2-py3-none-any.whl (21 kB)
Downloading urllib3-2.6.3-py3-none-any.whl (131 kB)
Downloading websocket_client-1.9.0-py3-none-any.whl (82 kB)
Downloading attrs-26.1.0-py3-none-any.whl (67 kB)
Downloading cffi-2.0.0-cp312-cp312-win_amd64.whl (183 kB)
Downloading outcome-1.3.0.post0-py2.py3-none-any.whl (10 kB)
Downloading PySocks-1.7.1-py3-none-any.whl (16 kB)
Downloading sniffio-1.3.1-py3-none-any.whl (10 kB)
Downloading wsproto-1.3.2-py3-none-any.whl (24 kB)
Downloading idna-3.11-py3-none-any.whl (71 kB)
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)

Installing collected packages: sortedcontainers, websocket-client, urllib3, sniffio, pysocks, pycparser, idna, h11, certifi, attrs, wsproto, outcome, cffi, trio, trio-websocket, selenium

WARNING: The script wsdump.exe is installed in
'C:\Users\kisha\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\LocalCache\local-packages\Python312\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

Successfully installed attrs-26.1.0 certifi-2026.2.25 cffi-2.0.0 h11-0.16.0 idna-3.11 outcome-1.3.0.post0 pycparser-3.0 pysocks-1.7.1 selenium-4.41.0 sniffio-1.3.1 sortedcontainers-2.4.0 trio-0.33.0 trio-websocket-0.12.2 urllib3-2.6.3 websocket-client-1.9.0 wsproto-1.3.2

[notice] A new release of pip is available: 25.0.1 -> 26.0.1

[notice] To update, run:

```
C:\Users\kisha\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip
```

```
C:\Users\kisha\selenium-lab>pip install webdriver-manager
```

Defaulting to user installation because normal site-packages is not writeable

Collecting webdriver-manager

Downloading webdriver_manager-4.0.2-py2.py3-none-any.whl.metadata (12 kB)

Collecting requests (from webdriver-manager)

Downloading requests-2.33.1-py3-none-any.whl.metadata (4.8 kB)

Collecting python-dotenv (from webdriver-manager)

Downloading python_dotenv-1.2.2-py3-none-any.whl.metadata (27 kB)

Collecting packaging (from webdriver-manager)

Downloading packaging-26.0-py3-none-any.whl.metadata (3.3 kB)

Collecting charset_normalizer<4,>=2 (from requests->webdriver-manager)

Downloading charset_normalizer-3.4.6-cp312-cp312-win_amd64.whl.metadata (41 kB)

Requirement already satisfied: idna<4,>=2.5 in

c:\users\kisha\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from requests->webdriver-manager) (3.11)

Requirement already satisfied: urllib3<3,>=1.26 in

c:\users\kisha\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from requests->webdriver-manager) (2.6.3)

Requirement already satisfied: certifi>=2023.5.7 in

c:\users\kisha\appdata\local\packages\pythonsoftwarefoundation.python.3.12_qbz5n2kfra8p0\localcache\local-packages\python312\site-packages (from requests->webdriver-manager) (2026.2.25)

Downloading webdriver_manager-4.0.2-py2.py3-none-any.whl (27 kB)

Downloading packaging-26.0-py3-none-any.whl (74 kB)

Downloading python_dotenv-1.2.2-py3-none-any.whl (22 kB)

Downloading requests-2.33.1-py3-none-any.whl (64 kB)

Downloading charset_normalizer-3.4.6-cp312-cp312-win_amd64.whl (154 kB)

Installing collected packages: python-dotenv, packaging, charset_normalizer, requests, webdriver-manager

WARNING: The script dotenv.exe is installed in

'C:\Users\kisha\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\LocalCache\local-packages\Python312\Scripts' which is not on PATH.

Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

WARNING: The script normalizer.exe is installed in

'C:\Users\kisha\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\LocalCache\local-packages\Python312\Scripts' which is not on PATH.

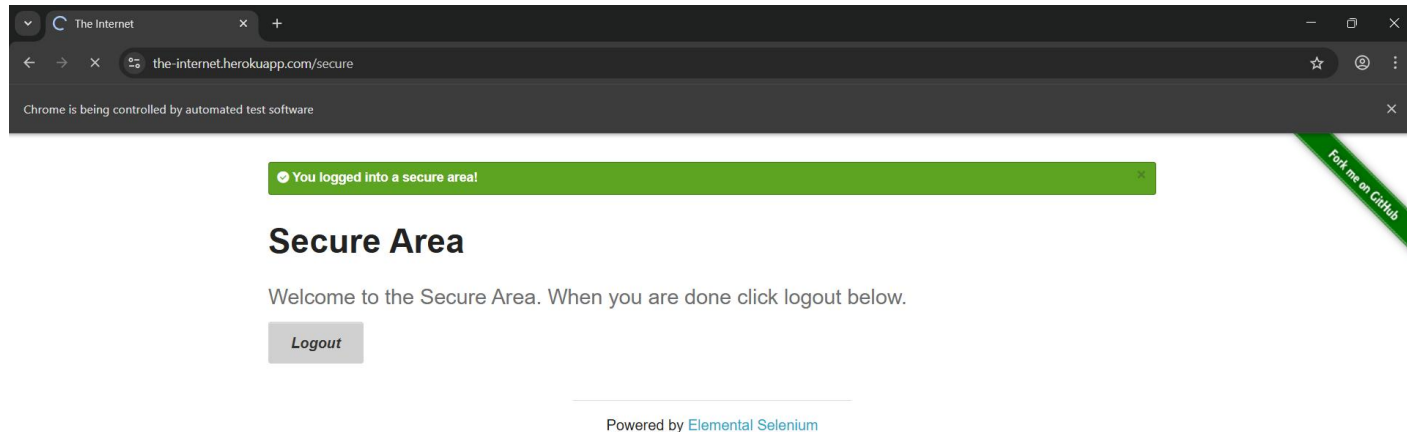
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.

Successfully installed charset_normalizer-3.4.6 packaging-26.0 python-dotenv-1.2.2 requests-2.33.1 webdriver-manager-4.0.2

[notice] A new release of pip is available: 25.0.1 -> 26.0.1

[notice] To update, run:

C:\Users\kisha\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip



```
C:\Users\kisha\selenium-lab>python selenium_login_test.py
Login Message: You logged into a secure area!
x
```

Task: Automate YouTube Search using Selenium:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from webdriver_manager.chrome import ChromeDriverManager
from selenium.webdriver.common.keys import Keys
import time

# Start browser
driver = webdriver.Chrome(service=Service(ChromeDriverManager().install()))

# Open YouTube
driver.get("https://www.youtube.com")
driver.maximize_window()

# Wait object
wait = WebDriverWait(driver, 10)

# Locate search box
search = wait.until(
    EC.presence_of_element_located((By.NAME, "search_query"))
)

# Enter text + press ENTER
search.send_keys("open source software")
search.send_keys(Keys.RETURN)

# Wait until results load
wait.until(
    EC.presence_of_element_located((By.ID, "video-title"))
)
print("Search results loaded successfully!")

# Wait 5 seconds to view results
time.sleep(5)

# Close browser
driver.quit()
```

Command Prompt:

```
C:\Users\kisha\selenium-lab>notepad youtube_search_test.py
```

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from webdriver_manager.chrome import ChromeDriverManager
from selenium.webdriver.common.keys import Keys
import time

# Start browser
driver = webdriver.Chrome(service=Service(ChromeDriverManager().install()))
# Open YouTube
driver.get("https://www.youtube.com")
driver.maximize_window()
# Wait object
wait = WebDriverWait(driver, 10)

# Locate search box
search = wait.until(
    EC.presence_of_element_located((By.NAME, "search_query"))
)
# Enter text + press ENTER
search.send_keys("open source software")
search.send_keys(Keys.RETURN)
# ✓ Wait until results load
wait.until(
    EC.presence_of_element_located((By.ID, "video-title"))
)
print("Search results loaded successfully!")
# Wait 5 seconds to view results
time.sleep(5)
# Close browser
driver.quit()
```

```
C:\Users\kisha\selenium-lab>python youtube_search_test.py
Search results loaded successfully!
```